

ARAP-Based Shape Editing to Manipulate the Center of Mass

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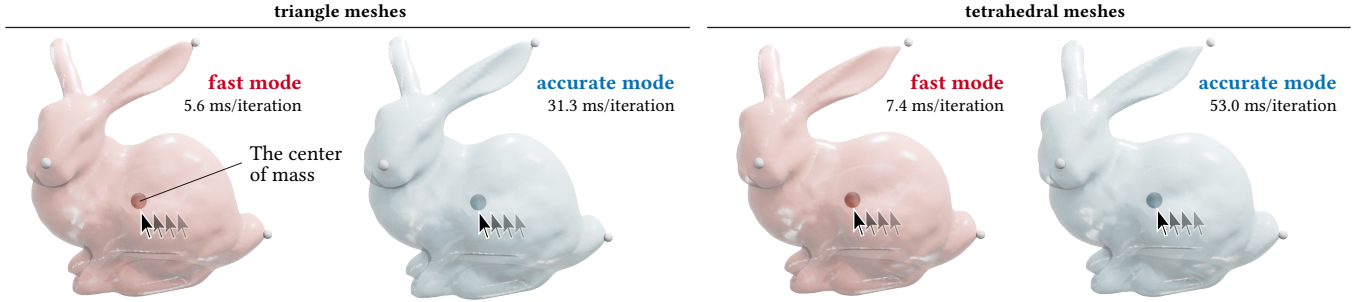


Figure 1: Our method enables users to interactively edit the center of mass of the input shape, just as the same as other handle points placed on the mesh. The input shape can be provided as either a triangulated surface mesh (left) or a tetrahedralized volume mesh (right). To accelerate the form-finding process, we offer two editing modes: *fast* and *accurate*.

Abstract

We propose an algorithm to align the center of mass of an input shape to a prescribed point. Unlike previous approaches that rely on hollowing the inner structure, our method deforms the outer shape by incorporating As-Rigid-As-Possible (ARAP) energy. Our method provides two editing modes: one prioritizing speed and the other focusing on accuracy. Combining these two modes, users can interactively explore the design space of shapes while achieving the desired center of mass positioning.

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1 Introduction

The center of mass is a fundamental physical quantity that determines the stability of objects. Controlling its position is critical in designing various industrial objects, such as tables, ceiling lights, toys, and more.

Over the past decade, computer graphics researchers have approached this problem as a topology optimization problem. Previous works have computed a hollowing structure inside a shape to align the center of mass with the target position [Bächer et al. 2014; Prévost et al. 2013]. Recently, Hafner et al. [2024] have even proved that the optimal hollowing structure always forms the shape of a quadratic function. However, when it comes to 3D printing these results, a hollowed shape must be printed in multiple parts due to the support removal and gluing process, which requires manual labor and reduces rigidity.

Thus, we present a different approach: deforming the outer shape so that the center of mass matches the given target position. We consider the center of mass when the volume is filled with a single material, which allows the user to 3D print the object in one piece. We built upon the As-Rigid-As-Possible (ARAP) shape deformation algorithm [Sorkine and Alexa 2007], and introduced simple equality constraints to account for the center of mass. Our framework

provides two editing modes, one prioritizing speed and the other accuracy.

2 Our Approach

Given the target center of mass position $g_* \in \mathbb{R}^3$ and an input closed triangle mesh or tetrahedral mesh denoted by vertices $V_0 \in \mathbb{R}^{n \times 3}$ and faces F (and tetrahedra T for a tetrahedral mesh), we aim to compute the new vertex positions $V_m \in \mathbb{R}^{n \times 3}$ such that the volumetric center of mass position matches g_* .

ARAP algorithm. Our method is built upon the ARAP deformation algorithm [Sorkine and Alexa 2007]. Given the vertex positions V_k at the k -th step, they minimize the *ARAP energy*

$$E_{\text{ARAP}}(V) = \text{tr}(V^T L V + V^T C R_k), \quad (1)$$

and obtain its next vertex positions

$$V_{k+1} = \underset{V}{\text{argmin}} E_{\text{ARAP}}(V), \quad (2)$$

where L is the discrete cotangent Laplacian, $C \in \mathbb{R}^{n \times 3n}$ is the constraint matrix, and $R_k \in \mathbb{R}^{3n \times 3}$ is the collection of rotation matrices (see [Sorkine and Alexa 2007] for details).

Computing the center of mass. The volumetric center of mass $g_k \in \mathbb{R}^3$ at the k -th step can be discretized as

$$g_k = \sum_{v_i, v_j, v_l \in F} \text{SVolume}(O v_i v_j v_l) \frac{1}{4} (v_i + v_j + v_l), \quad (3)$$

on a triangle surface mesh and

$$g_k = \sum_{t \in T} \text{SVolume}(t) \sum_{v_i \in t} \frac{v_i}{4}, \quad (4)$$

on a tetrahedral volumetric mesh, where $O \in \mathbb{R}^3$ is the origin, $O v_i v_j v_l$ is the tetrahedron formed by the four points, and $\text{SVolume}(\cdot)$ is the signed volume of the tetrahedron regarding the vertex permutation. In either case, g_k is described as

$$g_k = V_k^T m_k, \quad (5)$$

using a vector $m_k \in \mathbb{R}^n$.

Our algorithm. We make a simple modification to the ARAP deformation algorithm to solve

$$V_{k+1} = \underset{V}{\text{argmin}} E_{\text{ARAP}}(V), \quad (6)$$

$$\text{s.t. } V^T m_k = g_*, \quad (7)$$

at each step. By the KKT conditions, this can be done via a sparse linear solve

$$\begin{bmatrix} 2L & m_k \\ m_k^T & 0 \end{bmatrix} \begin{bmatrix} V \\ \lambda^T \end{bmatrix} = \begin{bmatrix} -C R_k \\ g_*^T \end{bmatrix}, \quad (8)$$

where $\lambda \in \mathbb{R}^3$ is the Lagrange multiplier. The computational cost is dominated by the factorization of the matrix on the left-hand side of Eq. 8.

This approach is already fast, but further speed-up can be achieved by simply fixing m_k to its value at the first step and using the same factorization across every step. The trade-off for the improved speed is a decrease in accuracy, leading to a larger deviation of the center of mass position from the prescribed g_* (see Figure 2 left). We refer

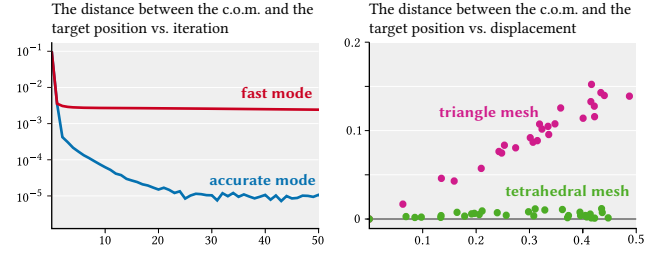


Figure 2: (left) The deviation between g_* and the center of mass stops decreasing after several iterations in our *fast* mode. **(right)** In our *fast* mode, as we displace g_* further from the initial shape's center of mass, the deviation between g_* and the center of mass after convergence becomes larger for triangle meshes, while it remains consistent for tetrahedral meshes.

to the two strategies as *accurate* and *fast*, and leave it to the users' choice.

3 Discussions and Conclusions

We implemented our code in C++ using LIBIGL, and tested it on both the triangulated (502 vertices, 1,000 faces) and tetrahedralized (1,156 vertices, 4,615 tetrahedra) Stanford Bunny. We also set several control handles on the mesh for ARAP shape manipulation, as shown by white dots in Figure 1.

As shown in Figure 1, the computation time per step for the *fast* mode was 5-8 times faster than the *accurate* mode. The drawback of the *fast* mode is that the center of mass has a larger deviation from the target position even after convergence (Figure 2 left), and this becomes worse when using a triangle mesh (Figure 2 right).

For future work, we plan to test our algorithm on a wider variety of meshes from various contexts. Also, developing an interactive design tool based on our two editing modes would be an interesting direction. The *fast* mode could be used to provide real-time feedback during clicking and dragging, while the *accurate* mode could be used when the user is finalizing their adjustments.

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